

- 182242

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- 182242

- Q.5 What is the unit of electrical power?
 a) Watt b) Joule
 c) Newton d) Tesla
- Q.6 Which instrument measures electric energy?
 a) Ammeter b) Voltmeter
 c) Energy meter d) Galvanometer
- Q.7 Which material is used for making fuses?
 a) Copper b) Tin-Lead alloy
 c) Aluminum d) Steel
- Q.8 The PIV of a diode refers to
 a) Peak Input Voltage
 b) Peak Inverse Voltage
 c) Peak Instantaneous Voltage
 d) None
- Q.9 Which device is used to improve power factor?
 a) Resistor b) Inductor
 c) Capacitor d) Transformer
- Q.10 The frequency of AC supply in India is:
 a) 50 Hz b) 60 Hz
 c) 100 Hz d) 25 Hz

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Power Factor.
 Q.12 Name the device used to improve power factor.
 Q.13 Write one application of Zener diode.

- Q.14 What is the frequency of AC supply in India?
 Q.15 Define energy and its unit.
 Q.16 What is the symbol of transformer in a circuit diagram?
 Q.17 What is meant by form factor in AC?
 Q.18 Name any two domestic electrical appliances that use induction motors.
 Q.19 What is a CVT?
 Q.20 Name any two instruments used for measuring electrical power.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 State Ohm's law and its mathematical expression.
 Q.22 Explain the difference between AC and DC with examples.
 Q.23 Write a short note on earthing and its importance.
 Q.24 State the applications of stepper motor.
 Q.25 Explain Faraday's Laws of Electromagnetic Induction.
 Q.26 List any three advantages of electrical energy over other forms of energy.
 Q.27 What is a star-delta starter? Explain briefly.
 Q.28 Describe the construction of a distribution system in domestic wiring.
 Q.29 Explain the working of a Zener diode as a voltage regulator.
 Q.30 Write short notes on different types of fuses.
 Q.31 State precautions for electrical safety.